

Evidence: four box cases

The blackout-box mystery

Name _____ Date _____

Constructed teaching cases. The same non-glowing white object stays inside.

A • Light sealed out

Torch off. No light enters. Object is not visible.

C • A small light leak

Torch off, but some room light enters. The object is faintly visible.

B • Torch lights the object

Torch on, aimed at the object. The viewing path is clear. Object is visible.

D • Wait in no light

No light enters. After one minute the object still cannot be seen.

W1. Compare A and B. What changed?

W2. Explain why C is different from D.

Route 1 • Find the light

Use cards A-D. Word bank: light / object / eye / darkness.

S1. In which case is the object clearly visible?

A. A

B. B

S2. Complete: We need ___ to see.

S3. Add arrows to show the order: source → object → ___.

S4. A place has a little light. Is it dim or completely dark?

A. Dim

B. Completely dark

S5. Explain why waiting does not make D visible.

Route 2 • Explain the evidence

Make a short record from the case cards.

Task	Case	Why is the view different?
T1	A	
T1	B	
T1	C	
T1	D	

T2. Draw and label a light path that lets us see the object.

T3. A pupil says the object disappears in A. Correct the claim.

Route 3 • Diagnose a leaky box

A team says its box has no light. They can still faintly see a white object.

X1. Explain why their observation challenges their no-light claim.

X2. Suggest one possible route by which light enters.

X3. Predict what happens if that route is fully blocked and no other light enters.

X4. Explain why identifying the object by touch is not evidence that it can be seen.

Exit and reflection

Point, draw, write or tell an adult.

E1. What do we need in order to see?

E2. Does light enter or leave the eye when we see?

E3. Can we see an object with no light at all?

R1. Which idea did you improve after hearing evidence?

R2. What would you investigate or ask next?
